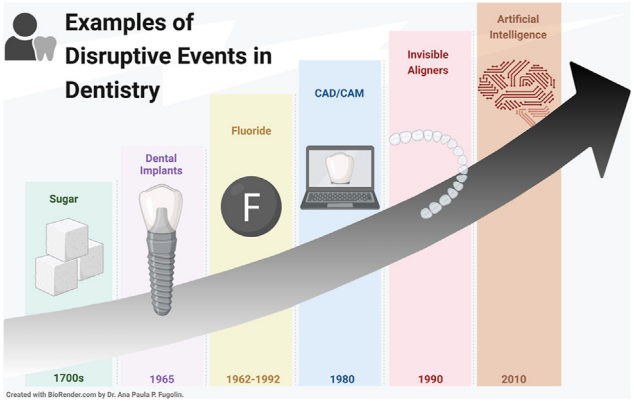
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Disruptive innovation events in dentistry

Jill Sirko, PhD, Wenyuan Shi, PhD

Editors' note: We are pleased to introduce a new series, "Innovations of Excellence, powered by ADA Forsyth," which provides a forum for brief introductions to new technologies, therapeutic approaches, and health care trends of interest to oral health clinicians and researchers—a "window" into the latest scientific discoveries. Inspired by the ADA's integration with The Forsyth Institute, now ADA Forsyth, JADA and JADA Foundational Science have partnered with ADA's new research arm to develop this series. These short articles are intended to raise awareness of emerging issues and cutting-edge research directions to keep the busy clinician apprised of exciting science trends expected to affect health care delivery and the practice of dentistry. Rather than describing the specifics of early or preliminary studies, however, these articles will present a more general view, highlighting specific innovations in dentistry. While ADA Forsyth will contribute significantly to the series, it is open to contributions from all potential authors and organizations with an oral health innovation to share. These reports will be published in both ADA journals, JADA and JADA FS. We hope you enjoy the topics and format of this new series and welcome thoughts and ideas you may have on topics you think would be useful to include in future issues.



Practical Implications

- Disruptive innovation events have shaped the field of dentistry throughout history.
- The authors explore how waves of scientific and technological innovation in oral health are driving a new era in dentistry.

Key Words. Disruptive innovation; oral health technology; dentistry; scientific innovation; technological advancements; artificial intelligence

We are in the middle of a technological revolution, accelerating innovation in science and technology every year. Global digitalization has completely changed the way people access, analyze, and share information.[1] Artificial intelligence (AI) is transforming how quickly we can synthesize information, make discoveries, and personalize patient care.[2]

The field of dentistry has been swept up in this wave of innovation, and suddenly, hundreds of new dental technology startup companies have emerged.[3,4] Ten years ago, that would have been hard to imagine. There is also an increasing number of investors focused on oral health, creating new opportunities for entrepreneurs and innovators to bring their ideas to market.[4] As a result, the dental market is growing exponentially.[5]

Disruptive innovation events have shaped the field of dentistry throughout history. Now, with the acceleration of new technologies and the imperative presented in the 2021 *Oral Health in America: Advances and Challenges* report[6] to harness those technologies to meet oral and systemic health needs globally, the dental field is poised for the next disruptive innovation event.

We introduce the idea of disruption as it relates to the field of dentistry. It is not a deep dive into particular innovations; rather, it sets the stage for future contributions exploring innovations that have the potential to disrupt and transform oral health care permanently. It is an invitation to learn more about these groundbreaking discoveries.

DISRUPTIVE INNOVATION AND DENTISTRY

The concept of disruptive innovation is not new. Introduced in a 1995 *Harvard Business Review* article by Bower and Christensen,[7] the theory of disruptive innovation explains how new services, technologies, and innovations can start small but eventually supplant established offerings in the market. The term disruptive has also been used more broadly to describe other types of events in human and scientific histories. Disruptive events can be natural disasters (eg, a hurricane) or human-caused catastrophes (eg, the atomic bomb).[8] They can be an accidental scientific discovery (eg, penicillin) or the introduction of new technology (eg, the internet).

Dentistry has been shaped by many notable disruptive events. For instance, the introduction of refined sugar to Western civilization was a disruptive event, causing a significant increase in caries rates and inciting the need for the dental profession as we know it.[9,10] Another such event was the discovery of the protective mechanism of fluoride and the resulting public health measure to fluoridate water in the United States. We call these

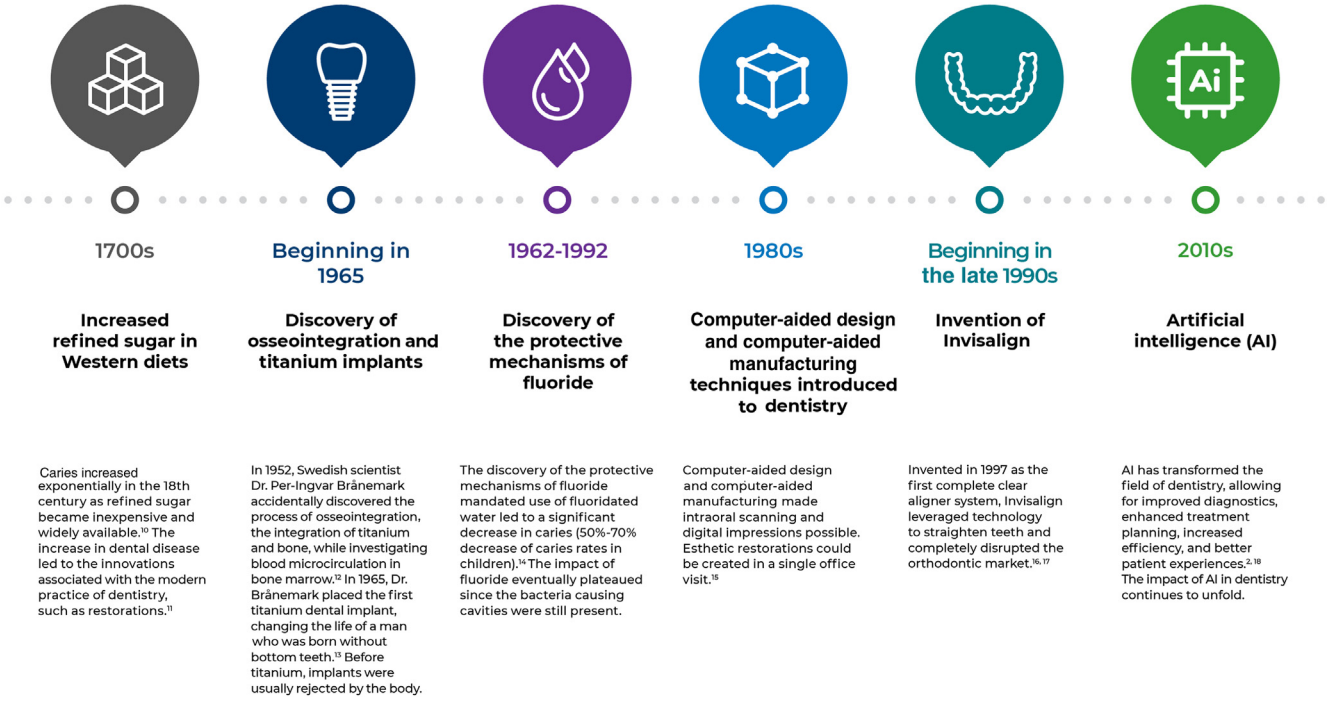


Figure. Disruptive innovation events and their impact on dentistry. These innovation events are disruptive because they permanently transformed the field. This list is not intended to be exhaustive.

events disruptive because they each caused a dramatic overhaul in the profession, permanently altering the dental landscape. The Figure provides a sampling of disruptive events, both scientific discoveries and disruptive technologies, that have impacted oral health treatment and the dental industry.

A NEW ERA IN DENTISTRY

There have been extraordinary technological, scientific, and medical advancements in the past 50 years. However, the dental profession has remained relatively unchanged. For instance, treatment for caries continues to rely on fluoride for prevention and restorations to address lesions. Yet there is a strong biological component—the local bacterial infection—that has yet to be addressed adequately. Caries are still the most common childhood disease.[19]

However, we are on the brink of a new era in the profession. We now understand there is a deep connection between oral health and systemic health.[20] Our mapping of the human oral microbiome means that the oral cavity is no longer a black box.[21,22] We know every species of bacteria that exists in the mouth.[23] We understand the mechanisms behind caries[24] and periodontal disease.[25-30] We have figured out how to use oral stem cells to regrow tissue.[31-34]

These rapid advancements in science and technology have primed the oral health field for the next disruptive innovation event.

POSSIBLE FUTURE DIRECTIONS

So, what is the next big thing in dentistry? Of course, we do not have a crystal ball, but there are a few key areas to watch.[6,35] Discoveries suggest important opportunities for incorporating effective biologically-based diagnostic tools in dentistry to predict both oral and systemic disease early, expanding dental preventative care beyond fluoride and hygiene.[6] With approximately 66% of the world now using the internet[36] and advancements in AI, the age of information is quickly becoming the age of intelligence.[37] The impact on oral health is already profound. AI-assisted digital dentistry and medicine-based technology will allow for personalized oral health care.[35] We likely will continue to see an increase in available diagnostic tools and new options for home-based care. Scientific innovation and discoveries in genomics, the oral microbiome, regenerative medicine, and biomaterials are creating improved treatment and prevention strategies. The next 20 years are likely to bring amazing new disruptive innovations with a marked positive impact on oral health care. ■

DISCLOSURE

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ADA Forsyth is actively developing various new technologies for managing dental diseases. This article does not endorse any of these products.

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